2(a)(i)	(component =) 4.8 sin 30° = 2.4 (N)	**A1**
2(a)(ii)	(8.2 × 0.50) **or** (W × 0.60) **or** (0.30 × 0.80) **or** (2.4 × 1.2)	**C1**
	(8.2 × 0.50) = (W × 0.60) + (0.30 × 0.80) + (2.4 × 1.2)	**C1**
	W = 1.6 N	**A1**
2(a)(iii)	force = 4.8 cos 30° = 4.2 N	**A1**
2(b)	$E = ½Fx$	**C1**
	0.32 = ½ × 8.2 × x x = 0.078 m	**A1**

Question	Answer	Marks
2(c)(i)	$((\Delta)E) = mg(\Delta)h$ **or** $W(\Delta)h$	**C1**
	$= 0.30 \times 0.090$ $= 0.027$ J	**A1**
2(c)(ii)	$E = ½mv^2$	**C1**
	$E = 0.044 – 0.027$ (= 0.017)	**C1**
	$v^2 = (2 \times 0.017) / (0.30 / 9.81)$ (where v = speed at A) $v = 1.1$ m s^{-1}	**A1**
	or	
	$E = ½mv^2$	**(C1)**
	$0.044 = ½ \times (0.30 / 9.81) \times v^2$ (where v = speed at B) ($v^2 = 2.88$) $v^2 = u^2 + 2as$ (where u = speed at A) $2.88 = u^2 + 2 \times 9.81 \times 0.090$	**(C1)**
	$u = 1.1$ m s^{-1}	**(A1)**
2(c)(iii)	(gravitational / resultant) force / weight is vertical	**B1**
2(c)(iv)	straight line with positive gradient starting from non-zero value of v_Y at time t_A to a time t_B	**B1**